Assignment 7 (25 points)

Arrays, Objects, Classes and Modules in JS

Task 1. - 10 points

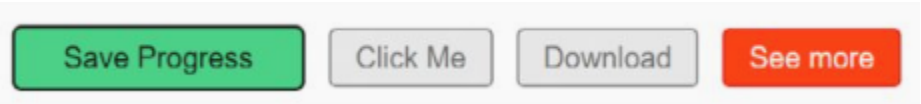
Create the class Button which has 3 initial properties: btnText, btnBgColor and btnTitle.

Define a method named "show" within this class, that render (display) the instance of the button by using document.write() method.

- btnText – used to specify the label on the button's instance;
- btnBgColor – the background color of the the button's instance;
- btnTitle - specifies extra information about the button's instance. This information is shown as a tooltip text when the mouse moves over the button's instance.

Create 2 arrays (arrTexts, arrColors) to store 4 items in each array.

Define the function to generate 4 button's instances like the following ones.



Use forEach() method to loop through 2 arrays (simultaneously) to create 4 buttons. BtnTitle is a composition of btnText and btnBgColor based on the template "*btnText* is shown on the *btnBgColor* background"

<u>Hint</u>: you need the following variation of forEach() method:

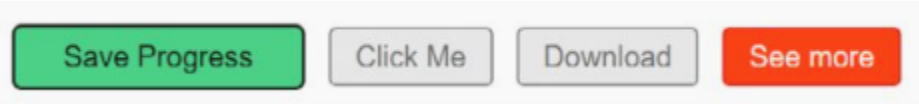
array1.forEach((element, index) {

... //element is an item of array1

.....//index is used to get the item from array 2 with the same index as an element

The created button's instances are stored in the arrButtons.

By using forEach() method loop through arrButtons array to display all buttons in the current document like shown below (define the function for this task):



Note!!! You can get an element (button) from an array after some period of time. For example: you get (display) the first button (Save Progress), wait for 30 seconds, next button (Click me), wait for 30 seconds - another button (Download), etc.

Hint: use forEach to loop the array with setTimeout () method not with setInterval().

Create the class ColorButton with one more property (fColor) which will inherit from the Button class. fColor is used to specify the font color of the button.

Override show() method to render the instance of the button based on 4 properties (btnText, btnBgColor, btnTitle, fColor).

Create and display one instance of the ColorButton.

Create the following modules to implement the above tasks:

- myButton.js
- myColorButton.js
- myArrays.js
- myFunctions.js
- Main.js

Task 2 – 15 points

There is a specified page template (you are not allowed to change it via HTML) - index.html.

```
<body>
  <div id="app">
    <p>Last news</p>
  <div id="content">
    <p></p>
    <p></p>
    <p></p>

  </div>
  <input id="myBtn" type="button" value="click me" />

  <script src="src/index1.js"></script>
</body>
```

Create a *News* class that creates (*render()* method) a block (*div* or *section* or *article*) consisting of

- a title,
- an image,
- a paragraph with the news text,
- a like counter (initially 0),
- a button LIKE (clicking on it corresponds to approving this news – increase like counter value) - use *onclick* event attribute to set the event handler (*incLikes()* method of the class)
- a button HIDE (clicking on it corresponds to "hiding" this news – an image the element will appear more transparent; a title and a paragraph will be displayed with a dark gray font color and a light gray background, a button LIKE will be disabled, the user can see it, but can't click) - use onclick event attribute to set the event handler (*hide()* method of the class)

The class has a method *show()* for displaying the news (which is a content created with *render()* method) in the specified block (method parameter). The number of "likes" is displayed as a corresponding number of "star"-symbols (you can use *♠*) below the title of the news.

Title1

Create a function *generatenews()* to select the paragraphs in the DOM page that are inside the *content* block (see the template above)

This function also creates news item (the instance of the *News* class) for each such paragraph and displays it inside that paragraph (via method show() of the instance).

Images recourses (string data for *src* attribute), titles and content for creating news are stored in the list of objects, like the following:

```
// Array of objects
let arrRecourses  = [
    {
        srcImg: 'Images/1.jpg',
        newsTitle: 'title1',
        newsContent: ' Lorem ipsum, or lipsum as it is sometimes known, is dummy
text used in laying out print, graphic or web designs. The passage is attributed
to an unknown typesetter in the 15th century who is thought to have scrambled
parts of Cicero's De Finibus Bonorum et Malorum for use in a type specimen book.
'
    },


    .....,


    {
       srcImg: 'Images/2.jpg',
        newsTitle: 'title2',
        newsContent: 'The purpose of lorem ipsum is to create a natural looking block of text
(sentence, paragraph, page, etc.) that doesn't distract from the layout. A practice not without
controversy, laying out pages with meaningless filler text can be very useful when the focus is meant to
```

be on design, not content.'

```
  }
];
```